own family. It works for all simple cases, but breaks down where incestuous mating occurs, and in certain insects, as we shall see.

First identify all the *common ancestors* of *A* and *B*. For instance, the common ancestors of a pair of first cousins are their shared grandfather and grandmother. Once you have found a common ancestor, it is of course logically true that all his ancestors are common to *A* and *B* as well. However, we ignore all but the most recent common ancestors. In this sense, first cousins have only two common ancestors. If *B* is a lineal descendant of *A*, for instance his great grandson, then *A* himself is the 'common ancestor' we are looking for.

Having located the common ancestor(s) of *A* and *B*, count the *generation distance* as follows. Starting at *A*, climb up the family tree until you hit a common ancestor, and then climb down again to *B*. The total number of steps up the tree and then down again is the generation distance. For instance, if *A* is *B*'s uncle, the generation distance is 3. The common ancestor is *A*'s father (say) and *B*'s grandfather. Starting at *A* you have to climb up one generation in order to hit the common ancestor. Then to get down to *B* you have to descend two generations on the other side. Therefore the generation distance is 1 + 2 = 3.

Having found the generation distance between *A* and *B* via a particular common ancestor, calculate that part of their relatedness for which that ancestor is responsible. To do this, multiply $\frac{1}{2}$ by itself once for each step of the generation distance. If the generation distance is 3, this means calculate $\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$ or $\left(\frac{1}{2}\right)^3$. If the generation distance via a particular ancestor is equal to *g* steps, the portion of relatedness due to that ancestor is $\left(\frac{1}{2}\right)^g$.

But this is only part of the relatedness between *A* and *B*. If they have more than one common ancestor we have to add on the equivalent figure for each ancestor. It is usually the case that the generation distance is the same for all common ancestors of a pair of individuals. Therefore, having worked out the relatedness between *A* and *B* due to any one of the ancestors, all you have to do in practice is to multiply by the number of ancestors. First cousins, for instance, have two common ancestors, and the generation distance via each one is 4. Therefore their relatedness is